

Université Sorbonne Paris Nord

École d'Ingénieurs Sup Galilée

Base de Données Avancées
TP 1

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Exercice 1

Définition de la table section

```
CREATE TABLE section (  
  course_id    VARCHAR(8),  
  sec_id       VARCHAR(8),  
  semester     VARCHAR(6) CHECK (semester IN ('Fall', 'Winter', 'Spring',  
    ↪ 'Summer')),  
  year         NUMERIC(4,0),  
  building     VARCHAR(15),  
  room_number  VARCHAR(7),  
  time_slot_id VARCHAR(4),  
  PRIMARY KEY (course_id, sec_id, semester, year),  
  FOREIGN KEY (course_id) REFERENCES course,  
  FOREIGN KEY (building, room_number) REFERENCES classroom  
);
```

Insertion du cours BIO-101

```
INSERT INTO course VALUES ('BIO-101', 'Intro. to Biology', 'Biology', '4');
```

Exercise 2

```
-- 1
DESC section;
SELECT * FROM section;

-- 2
SELECT * FROM course;

-- 3
SELECT title, dept_name
FROM course;

-- 4
SELECT dept_name, budget
FROM department;

-- 5
SELECT name, dept_name
FROM teacher;

-- 6
SELECT name
FROM teacher
WHERE salary > 65000;

-- 7
SELECT name
FROM teacher
WHERE salary BETWEEN 55000 AND 85000;

-- 8
SELECT DISTINCT dept_name
FROM teacher;

-- 9
SELECT name
FROM teacher
WHERE dept_name = 'Comp. Sci.'
    AND salary > 65000;

-- 10
SELECT *
FROM section
WHERE semester = 'Spring'
    AND year = 2010;

-- 11
SELECT title
FROM course
WHERE dept_name = 'Comp. Sci.'
```

```

    AND credits > 3;

-- 12
SELECT t.name, t.dept_name, d.building
FROM teacher t
JOIN department d ON t.dept_name = d.dept_name;

-- 13
SELECT DISTINCT s.name
FROM student s
JOIN takes t ON s.ID = t.ID
JOIN course c ON t.course_id = c.course_id
WHERE c.dept_name = 'Comp. Sci.';

-- 14
SELECT DISTINCT s.name
FROM student s
JOIN takes t ON s.ID = t.ID
JOIN teaches te ON t.course_id = te.course_id
                AND t.sec_id = te.sec_id
                AND t.semester = te.semester
                AND t.year = te.year
JOIN teacher i ON te.ID = i.ID
WHERE i.name = 'Einstein';

-- 15
SELECT DISTINCT course_id, ID
FROM teaches;

-- 16
SELECT course_id, sec_id, COUNT(*)
FROM takes
WHERE semester = 'Spring' AND year = 2010
GROUP BY course_id, sec_id;

-- 17
SELECT dept_name, MAX(salary)
FROM teacher
GROUP BY dept_name;

-- 18
SELECT course_id, sec_id, semester, year, COUNT(*)
FROM takes
GROUP BY course_id, sec_id, semester, year;

-- 19
SELECT building, COUNT(*)
FROM section
WHERE (semester = 'Fall' AND year = 2009)
     OR (semester = 'Spring' AND year = 2010)
GROUP BY building;

```

```

-- 20
SELECT d.dept_name, COUNT(*)
FROM course c
JOIN department d ON c.dept_name = d.dept_name
JOIN section s ON c.course_id = s.course_id
WHERE s.building = d.building
GROUP BY d.dept_name;

-- 21
SELECT DISTINCT c.title, t.ID
FROM teaches t
JOIN course c ON t.course_id = c.course_id;

-- 22
SELECT semester, COUNT(*)
FROM section
WHERE semester IN ('Summer', 'Fall', 'Spring')
GROUP BY semester;

-- 23
SELECT s.ID, s.name, SUM(c.credits) AS total_credits
FROM student s
JOIN takes t ON s.ID = t.ID
JOIN course c ON t.course_id = c.course_id
WHERE s.dept_name <> c.dept_name
GROUP BY s.ID, s.name;

-- 24
SELECT d.dept_name, SUM(c.credits) AS total_credits
FROM department d
JOIN course c ON d.dept_name = c.dept_name
JOIN section s ON c.course_id = s.course_id
GROUP BY d.dept_name;

```